

Ash Zhang

Full Stack Software Engineer

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SUMMARY

Experienced and dedicated Software Engineer with 9+ years experience in developing, designing, and maintaining sophisticated websites and web applications. Strong customer-facing and communication skills, with the ability to translate complex technical concepts into customer-friendly solutions. Specializes in Front-end and Computer Graphics technique. ACM SIGGRAPH Member. Highly enthusiastic about embracing new advancements in the field, consistently seeking opportunities to expand knowledge and skills.. Regular attendee of programmer hackathons.

Skills

Web development

- JavaScript · Java · Python · C# · Solidity · HTML5 & CSS3 · SASS · SQL
React.js · Vue.js · Three.js · Babylon.js · Cesium.js · Gatsby
Node.js · Express.js · Flask · jQuery
Web3 · ERC20 · Hardhat · Truffle
MongoDB · PostgreSQL · MySQL · GraphQL · Bootstrap · Ant-design · Material UI
React-Query · Redux · Axios · Firebase · Vuex
Git · Webpack · Babel · Mocha · Jest · AWS

Professional Experience

09/2023 – present

Software Engineer, *Komatsu Mining*

- Engineered enterprise-grade **3D Digital Twin** visualization software for autonomous mining operations.
- Developed real-time **Computer Graphics** systems using BabylonJS, WebGL, React, and TypeScript.
- Designed interactive Guidance Systems for operator visualization and situational awareness.
- Resolved complex engineering challenges across rendering performance, **GPU** efficiency, **memory management**, and application scalability.
- Built reusable visualization frameworks and interactive **3D** user experiences.
- Partnered with product managers, UX designers, and field specialists to translate **operational requirements into practical software solutions**.
- Supported customer **demonstrations**, feature showcases, and technical discussions by presenting product capabilities and explaining design decisions to **internal and external stakeholders**.
- Contributed to product strategy by **evaluating customer feedback, identifying usability improvements**, and influencing feature prioritization for industrial visualization workflows.

07/2021 – 09/2023

Full Stack Engineer, *Mountain Pass Pty Ltd*,

- Optimized and developed multiple **React.js** Web applications based on wireframe prototypes.
- Implemented and maintained **Restful APIs** for seamless cooperation between FrontEnd and BackEnd development.

- Build **GIS** application from scratch using **Cesium.js** and **Resium.js**.
- Created a **Dapp** website, including wrote ERC20 token smart contract using **solidity**, and built the **Web3** website.
- Developed an experimental **Babylon.js** 3D Web Application and integrated with Metamask. Solved the **Three.js** issues for the team.
- Implemented **React-Query** State management interfaces and facilitated communication with the BackEnd using Axios.
- Contributed to the **open-source** community by debugging libraries and submitting pull requests.
- Engaged in regular meetings with clients and product managers to discuss and propose multiple solutions for system enhancements.
- Provided guidance and mentorship to junior developers, aiding them in debugging and troubleshooting during development. Built the team **WIKI** and host hosted informative tech sharing sessions on **Three.js**,
- Enhanced user interfaces by encapsulating React components with React-Bootstrap and customizing styling and functionality.

12/2021 – 04/2022

Full Stack Engineer (Casual), University of Technology, Sydney

- Developed a user-friendly web GUI using **React.js** for front-end and **Flask** for back-end, enabling the execution and creation of quantum computing models.
- Implemented innovative solutions for **drag and drop** functionality within a grid layout, enhancing the user experience and interaction.
- Collaborated closely with the research team to design and develop intuitive features, ensuring seamless integration of the GUI with the quantum computing models.

03/2017 – 04/2018

Web Developer Programmer, Tap4fun Co., Ltd,

- Designed and developed **5** web applications and websites for internal clients. Using **Vue.js, React.js, Flask, Node.js, Bulma, MongoDB** to build MVC architecture websites with RESTful APIs.
- Created a set of Vue.js components for the internal framework. Increased the efficiency of developing and rapid prototyping for the web development team.
- Developed unit testing for each project for the features testing and data validations.
- Designed and keep optimized a user-friendly backstage management portal for internal websites.
- Hosted 2 technology-sharing sessions about React.js and Vue.js for a 30-people Web development team.

Education

02/2020 – 11/2021

Master of Information Technology,

University of Technology Sydney, Faculty of Engineering and Information Technology
High Distinction. GPA 6.6 out of 7. WAM 87
2022/2021 Postgraduate Dean's list

09/2013 – 06/2017

Bachelor of Engineering, Electronic Engineering,

University of Electronic Science and Technology of China
The First Price Scholarship

Side Projects

01/2023 – 03/2024	PaaS application for the natural resources GIS system Designed and integrated GIS functionalities into the application base on Cesium.js and React.js, allowing users to effectively analyze and manage natural resources data on 3D map.
08/2021 – 11/2021	Covid-3D online visualisation dashboard ↗ , <i>(Research project)</i> - Used three.js and React.js to create an online dashboard for Covid-19 3D visualization - Retrieved data from Public Covid API and made the data cleaning
04/2021 – 10/2021	Voronoi structure generating tool , <i>(Research project)</i> - Used three.js as the rendering engine. Increased the performance of rendering Voronoi structure - Implementing Delaunay triangulation and Jump Flood Algorithm in the geometry computing process. Researched the approach to store distance fields and applied the ray-marching technique to generate the structure.
03/2021 – 03/2021	Virtual Museum Experience ↗ , <i>(Hackathon project)</i> - Developed an online web application using three.js and vue.js. - Developed an interactive virtual museum web application in a two-day Hackathon. Developed a virtual scene that includes a complex light environment and multiple cameras. Created a GUI to control the setting of the environment. - Created a tool to load 3D objects into the scene using JSON configurations. Made the project scalable to connect to other scenes after finishing the Hackathon.

Certificates

Introduction to HTML5 ↗ University of Michigan	Interactivity with JavaScript ↗ University of Michigan
Advanced Styling with Responsive Design ↗ University of Michigan	Introduction to CSS3 ↗ University of Michigan
Server-side Development with NodeJS, Express and MongoDB ↗ The Hong Kong University of Science and Technology	Azure AI Fundamentals ↗ Microsoft Azure

Awards

04/2022	2022 Postgraduate Dean's list , <i>University of Technology, Sydney</i>
04/2022	UTS Techcelerator 2021 Top projects , <i>University of Technology, Sydney</i>
06/2021	2021 Student Games Showcase , <i>UTS Games Studio</i>
04/2021	2021 Postgraduate Dean's list , <i>University of Technology, Sydney</i>